Imaging-based single-cell spatial data: Xenium

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Day 2 – 17.06.2026

Recap and today

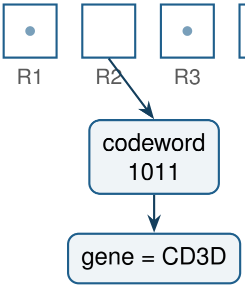
Yesterday: spot-based Visium, the standard workflow, and first spatial statistics.

Today we move to **imaging-based** data at single-cell resolution:

- how the data are produced and why segmentation is the key step;
- quality control and evaluation metrics specific to this technology;
- a full single-cell spatial workflow, and the method choices along the way.

How imaging-based ST works

- A targeted gene panel is detected *in situ* by cyclic hybridisation and imaging of fluorescent probes.
- Each gene has a code; across rounds the code is read out and decoded to a transcript at an (x, y) location.
- Transcripts are assigned to cells using a segmentation mask.
- Output: transcripts, a cell $\times$ gene matrix, cell metadata, and images.



Imaging-based platforms

Platform	Panel size	Note
MERFISH / Vizgen	$\sim$300–1000	error-robust barcodes
seqFISH+	$\sim$10,000	many pseudocolours
Xenium (10x)	280–480, up to 5000 (Prime)	in situ, H&E compatible
CosMx (NanoString)	$\sim$1000–6000	RNA and protein

We use Xenium with the 377-gene Multi-Tissue and Cancer panel. The panel choice bounds what biology you can see, so it is part of the experimental design.

Xenium outputs and how to read them

- `cell_feature_matrix.h5`: cell $\times$ feature counts (genes and controls).
- `cells.parquet`: centroid, area, counts per cell.
- `transcripts.parquet`: every decoded transcript (large).
- `morphology_*.ome.tif`: images (large).

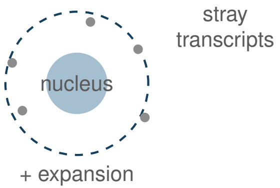
Two reading paths. `spatialdata_io.xenium` builds a full object with images and shapes (great for visualisation, heavy). For table-level analysis we read the matrix and cell table straight into AnnData. Same cells, a fraction of the memory: this is what keeps us inside 25 GB.

Segmentation is the crux

Cells are defined by a mask, and everything downstream inherits its errors.

- In dense tissue, boundaries are ambiguous: transcripts are misassigned, and small or nucleus-less cells are missed.
- Symptoms: implausible co-expression, “cells” with too few transcripts, lineage markers bleeding across boundaries.

Treat cell-level results as estimates and sanity-check against the raw transcript image where it matters.



Segmentation methods

- **Default (10x):** nuclear stain plus $\sim$15 µm expansion; a multimodal deep-learning kit uses membrane/boundary stains when available.
- **Image-based:** Cellpose (generalist nucleus/membrane segmentation).
- **Transcript-based:** Baysor and Proseg learn transcript colocalisation; Proseg uses a cellular Potts model.
- **Learned, reference-aware:** segger (graph neural network, transcript to cell link prediction, can use scRNA-seq).

Benchmarks show a tradeoff between transcript capture and cell purity that is tissue-dependent; there is no universal winner.

Stringer et al., *Nat Methods* 2021 (Cellpose); Petukhov et al. (Baysor); Jones et al. 2025 (Proseg); Heidari et al. 2025 (segger).

Why segmentation matters for your biology

- Misassigned transcripts directly corrupt cell typing and, especially, cell–cell communication (a ligand can “jump” to the wrong cell).
- Nucleus-anchored methods miss cells without a captured nucleus, biasing composition.
- Improving identity in poorly segmented, dense regions is an active and unsolved problem.

[figure slot]

the same field of view segmented by two methods, with cell counts and a marker that should not co-occur

suggested source: a segmentation-benchmark figure (e.g. segger or CRISP), or your own

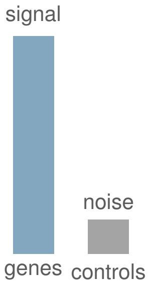
Heidari et al. 2025 (segger); CRISP segmentation benchmark, bioRxiv 2026.

Control features

The panel ships with built-in negative controls:

- **Negative control probes**: target no real transcript; estimate non-specific binding.
- **Negative control codewords (blanks)**: valid codes assigned to no gene; estimate the decoding error rate.

A healthy run puts only a small fraction of counts on controls. A high control fraction in a cell or region is a warning.



Quality control and evaluation metrics

- **Transcripts / cell** and **genes / cell**: too few means an empty or fragmented segment.
- **Control fraction / cell**: flags noise.
- **Cell area**: very small or very large segments are suspect.
- **Counts vs area**: density should be roughly consistent.
- **Panel sensitivity**: counts are far below scRNA-seq, so a gene's absence is weak evidence.

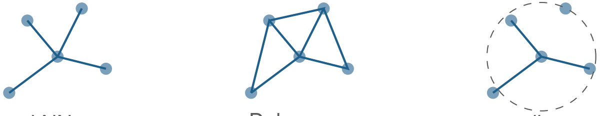
Read every metric in space, not only as a histogram.

Cell typing on a small panel

- Fewer genes make clusters noisier and rare types harder to separate.
- Use a curated marker set for the tissue, then refine by eye and check spatial plausibility.
- Reference label transfer (CellTypist, scANVI, `ingest`) scales when a matched scRNA-seq reference exists, but inherits any segmentation contamination.
- BANKSY can type cells and find domains together using the neighbourhood.

Domínguez Conde et al., *Science* 2022 (CellTypist); Singhal et al., *Nat Genet* 2024 (BANKSY).

Choosing a spatial graph



kNN fixes the neighbour count; Delaunay follows tissue geometry; radius fixes a physical distance. The choice changes enrichment results, so state it.

Single-cell spatial statistics

- **Neighbourhood enrichment**: which cell types co-locate vs avoid each other (permutation test on the graph).
- **Co-occurrence**: how that association changes with distance, separating tight contact from regional association.
- **Ripley's K / L**: clustering vs dispersion of a cell type across scales.
- **Spatially variable genes**: Moran's I over cells.

These are the building blocks for the organisation analysis on Day 3.

Beyond one section

- Multiple sections bring batch effects. Integration with Harmony or scVI aligns embeddings before joint clustering.
- Spatial-aware methods (e.g. BANKSY) can do batch-aware domain detection.
- Comparing conditions needs composition-aware tests, not just eyeballing proportions (Day 3).

Korsunsky et al., *Nat Methods* 2019 (Harmony); Lopez et al., *Nat Methods* 2018 (scVI).

Open problems on imaging-based data

- Segmentation in dense tissue, and its propagation into typing and communication.
- Low detection of cytokines and chemokines, so immune states are hard to identify.
- Reliability of spatial QC metrics, which are still being standardised.
- Integrating images (H&E) with the molecular layer for better region identity.

Further reading

- Petukhov et al. Baysor: cell segmentation from transcripts.
- Jones et al. 2025 (Proseg); Heidari et al. 2025 (segger); Stringer et al., *Nat Methods* 2021 (Cellpose).
- Singhal et al. BANKSY. *Nat Genet* 2024.
- 10x Genomics analysis guides for Xenium segmentation and QC.
- Squidpy tutorials for Xenium: `squidpy.readthedocs.io`.

Hands-on: `day_2/02_tutorial.ipynb`

Dataset: a public Xenium non-diseased human kidney section ($\sim$97,560 cells).
You will:

- read the section the lean way (matrix + cell table);
- separate control features and compute Xenium QC;
- filter, normalise, cluster, and annotate cell types with markers;
- plot cell types in space; build a spatial graph and run enrichment, co-occurrence, and Moran's I.

Exercises: set thresholds and map removed cells, refine annotation, compare kNN vs Delaunay graphs, and interpret spatially variable genes.